

DIGITALLY STAMPED HEAT COMPLIANCE CERTIFICATE

Specific Energy Consumption (SEC) and Carbon Border Adjustment Mechanism (CBAM) Audit

Heat Serial Number:	HEAT-1042	Verification Timestamp:	2026-09-26 19:07:06
Manufacturing Facility:	Kolhapur Foundry Cluster Unit #14	Edge Gateway ID:	SE-ECO-EDGE-4102
Energy Metering:	Schneider Electric EasyLogic PM5350	Furnace Type:	1.5 MT Coreless Medium-Freq Induction
Weighbridge Net Output:	1.500 Metric Tonnes	Metered Ingot Energy:	937.5 kWh

STATUTORY MEASUREMENT AND AUDIT METHODOLOGY

Specific Energy Consumption (SEC) = Metered Total kWh / Weighbridge Net Tonnes = 937.5 / 1.500 = 625.0 kWh/tonne

Embodied Scope 2 Carbon = SEC x 0.82 kg CO2/kWh (CEA India Baseline) = 625.0 x 0.82 = 512.5 kg CO2/tonne

Audit Dimension	Measured Heat Value	Statutory Benchmark	Compliance and Economic Result
Specific Energy (SEC)	625.0 kWh/tonne	BEE Target: 625.0 kWh/t Kolhapur Avg: 780.0 kWh/t	VERIFIED (Delta: +0.0 kWh/t)
Embodied Carbon Intensity	0.712 tCO2/tonne	EU CBAM Target: 1.50 tCO2/t India Benchmark: 2.50 tCO2/t	COMPLIANT
EU CBAM Duty Liability	EUR 0.00 / tonne (INR 0 / tonne)	EU ETS Allowance Rate: EUR 79.68 / tonne CO2	+ EUR 79.68/t saved (+ INR 7450/t export shield)
BEE PAT Scheme ESCerts	0.0290 ESCerts	EC Act 2001 (BEE PAT Cycle) 1 ESCert = 1 Mtoe Saved	INR 62 IEX Market Value

DIGITAL AUDIT VERIFICATION SEAL

Schneider Electric EcoStruxure(TM) Edge Gateway Cryptographic Certificate
Verified under ISO 50001 (Energy Management) and GHG Protocol Scope 2 Guidelines.
SHA-256 Stamp: f06c2a9284057fc49e92fec019e71da1d721e25b8d0415b74d0d1cecd79283bb

AUTHORIZED DIGITAL SEAL
[DIGITALLY SIGNED]

Node: SE-ECO-EDGE-4102
Status: VALID & IMMUTABLE